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37. $K_C = \dfrac{[NH_3]^4[O_2]^5}{[NO]^4[H_2O]^6}$

Chemical equation is,

$$4\,NO + 6H_3O \rightleftharpoons 4NH_3 + 5O_2$$

40. Given that $[PCl_5]_{initial} = \dfrac{1 mole}{1\,dm^3}$

$[Cl_2]_{eq} = 0.6$ mole dm^{-3}

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

$[PCl_3]_{eq} = 0.6$ mole dm^{-3}

$[PCl_5]_{eq} = 0.4$ mole dm^{-3}

$$\therefore K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{0.6 \times 0.6}{0.4}$$

$K_C = 0.9$

41. for the reaction,

$SrCO_3\,(S) \rightleftharpoons SrO(S) + CO_2(S)$

$\Delta ng = 1 - 0 = 1$

$\therefore K_P = K_C\,(RT)$

$2.2 \times 10^{-4} = K_C\,(0.0821)\,(1002)$

$$K_C = \frac{2.2 \times 10^{-4}}{0.0821 \times 1002}$$

$$= 2.674 \times 10^{-6}$$

42. V = 3L

$$[HI]_{initial} = \frac{0.3\ mol}{3L} = 0.1\ M$$

$[HI]_{eq} = 0.05$ M

$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

	HI(g)	$H_2(g)$	$I_2(g)$
Initial Concentration	0.1	–	–
Reacted	0.05	–	–
Equilibrium concentration	0.05	0.025	0.025

$$K_C = \frac{[H_2][I_2]}{[HI]^2}$$

$$= \frac{0.025 \times 0.025}{0.05 \times 0.05}$$

$K_C = 0.25$

$K_P = K_C\,(RT)^{\Delta ng}$

$\Delta ng = 2 - 2 = 0$

$K_P = 0.25\,(RT)^o$

$K_P = 0.25$

43. $2NO\,(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

	NO	O_2	NO_2
Initial pressure	1	1	–
Reacted	0.96	0.48	–
Equilibrium partial pressure	0.04	0.52	0.96

$$K_P = \frac{\left(P_{NO_2}\right)^2}{\left(P_{NO}\right)^2\left(P_{O_2}\right)} = \frac{0.96 \times 0.96}{0.04 \times 0.04 \times 0.52}$$

$K_P = 1.017 \times 10^3$

44. $CH_4(g) + 2H_2S(g) \rightleftharpoons CS_2(g) + 4H_2(g)$

$K_C = 4 \times 10^{-2}$ mol lit^{-2}

Volume = 500 ml = $^1/_2$ L

$$[CH_4]_{in} = \frac{1\ mol}{1/2\ L} = 2\ mol\ L^{-1} \qquad [CS_2]_{in} = \frac{1\ mol}{1/2\ L} = 2\ mol\ L^{-1}$$

$$[H_2S]_{in} = \frac{2\ mol}{1/2\ L} = 4\ mol\ L^{-1} \qquad [H_2] = \frac{2\ mol}{1/2\ L} = 4\ mol\ L^{-1}$$

$$Q = \frac{[CS_2][H_2]^4}{[CH_4][H_2S]^2}$$

$$\therefore Q = \frac{2 \times (4)^4}{(2) \times (4)^2} = 16$$

$Q > K_C$

$\therefore$ The reaction will proceed in the reverse direction to reach the equilibrium.